

Michelle Herrera-Cuen

Computer Engineering Student | Embedded Systems | Robotics | Firmware | Control Systems

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Technical Skills: C • C++ • Python • Verilog • Microcontrollers • LoRa Git • Linux • VS Code • ESP-IDF • Altium Designer

Education

Bachelor of Science in Computer Engineering

California State University, Long Beach | Expected 2027

Experience

Honors Thesis: Robotics Firmware

Undergraduate Research Assistant – Long Beach, CA | Jan 2025 - Sep 2025

- Developed a configurable robotics ecosystem enabling firmware generation for ESP32, Teensy, STM32, and Arduino platforms; allowing no-code robot configuration through desktop software.

Jack in the Box

Shift Lead – Torrance, CA | Jan 2022 - Present

- Develop leadership, communication, and time management skills in a high-pressure environment.

Engineering Projects

Embedded Space Invaders Game

- Developed a real-time embedded game in C using the TM4C123 LaunchPad.
- Integrated Nokia 5110 LCD graphics, ADC joystick input, GPIO interrupts, and PWM audio output.
- Optimized firmware for responsive gameplay under embedded hardware constraints.

Autonomous Mobile Robots

- Designed object-following and wall-following robots using infrared sensors and DC motors.
- Implemented PID-style control logic and obstacle detection.
- Integrated UART, SPI, ADC, PWM, GPIO interrupts, and motor drivers.

Interactive Audio Reaction FPGA Game

- Designed digital hardware in Verilog on the Nexys A7 FPGA.
- Implemented finite state machines, PWM audio generation, LFSRs, and seven-segment displays.
- Verified functionality using Vivado simulation and hardware testing.

IoT Machine Learning Pipeline

- Built an IoT analytics pipeline using simulated smart appliance data.
- Processed sensor telemetry from PostgreSQL/NeonDB.
- Evaluated Linear Regression, KNN, and SVM models.

Leadership

Society of Automotive Engineers

Data Acquisition Team Lead | Jan 2024 - Sep 2025

- Developed an ESP32-based telemetry and data acquisition system for Formula SAE competition vehicles.
- Implemented LoRa communication between ESP32 and Raspberry Pi for real-time sensor data collection.
- Designed firmware to log suspension, IMU, and drivetrain data in real time.

Substack Writer & Creator

missingmichelle.substack.com | Jan 2026 - Present

- Publish weekly essays exploring personal growth, culture, and life experiences
- Demonstrate consistent long-form writing, storytelling, and audience engagement through an independent publication

Women in Computing

Outreach Team | Jan 2025 - Present

- Hosted a workshop to guide young women interested in STEM on the basics of HTML and CSS language